

MSD, VACF and $g(r)$ in Practice

Tutors:

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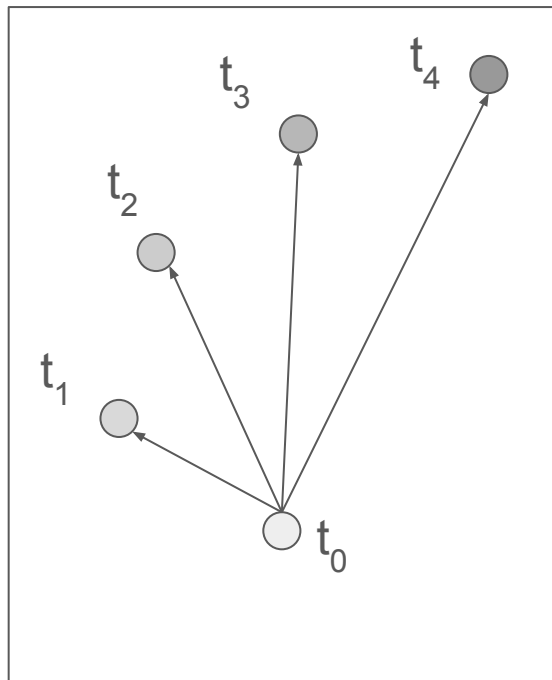
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Federica Troni

Mean Square Displacement (MSD): Definition

$$MSD(t) = \langle |\mathbf{r}(t) - \mathbf{r}(0)|^2 \rangle_{\text{atoms}}$$

MSD: Compute $MSD(t) = \langle |\mathbf{r}(t) - \mathbf{r}(0)|^2 \rangle_{\text{atoms}}$



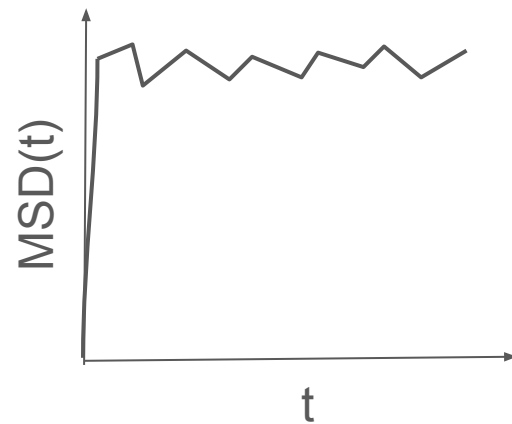
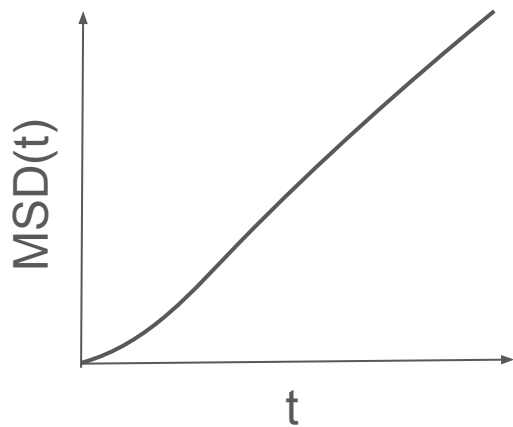
Single Particle

traj			
t ₀	1, 1, 1		
	3, 2, 1		
	3, 2, 1		
t ₁	2, 3, 2		
	1, 1, 0		
	4, 1, 2		

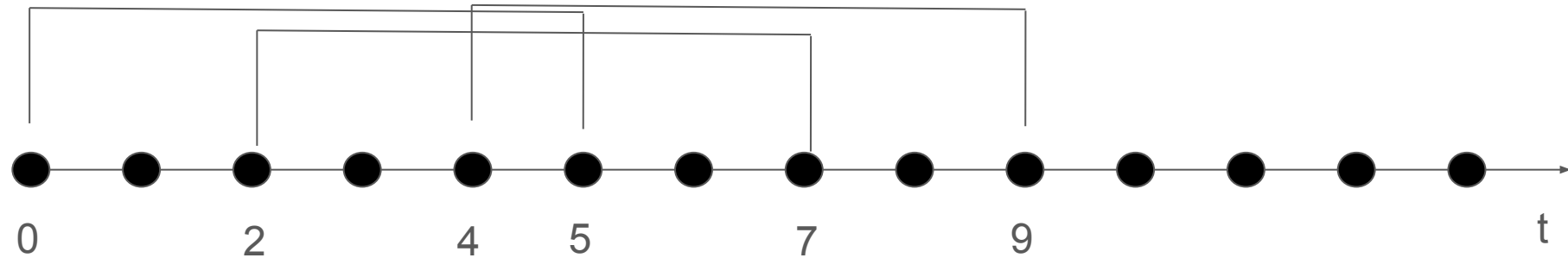
Three Particles

#atoms
#frames
#dims
traj.shape = (2,3,3)

MSD: Discussion



Equilibrium



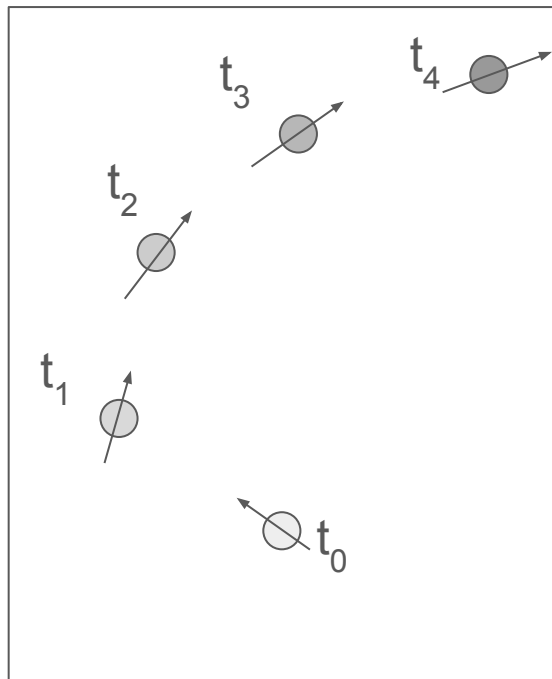
In equilibrium, statistical properties do not depend on the initial configuration

Velocity Autocorrelation Function (VACF): Definition

$$\text{VACF}(\tau) = \langle \boldsymbol{v}(0) \cdot \boldsymbol{v}(\tau) \rangle_{\text{atoms, time}}$$

$$\text{VACF}(\tau) = \langle \mathbf{v}(0) \cdot \mathbf{v}(\tau) \rangle_{\text{atoms, time}}$$

VACF: Compute



Single Particle

t_0	1, 1, 1
	3, 2, 1
	3, 2, 1

t_1	2, 3, 2
	1, 1, 0
	4, 1, 2

t_3	3, 2, 1
	2, 1, 0
	3, 2, 2

Three Particles

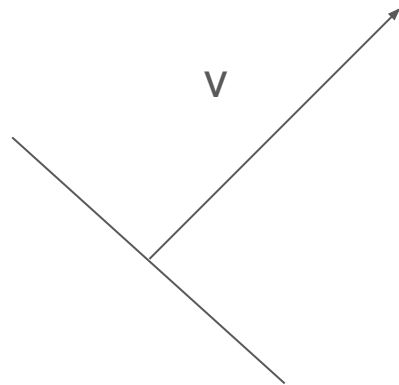
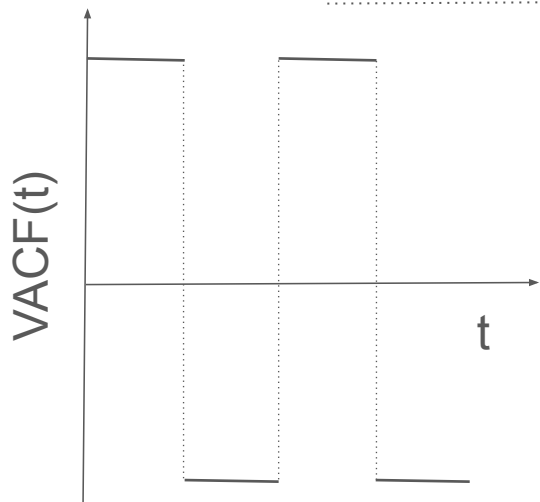
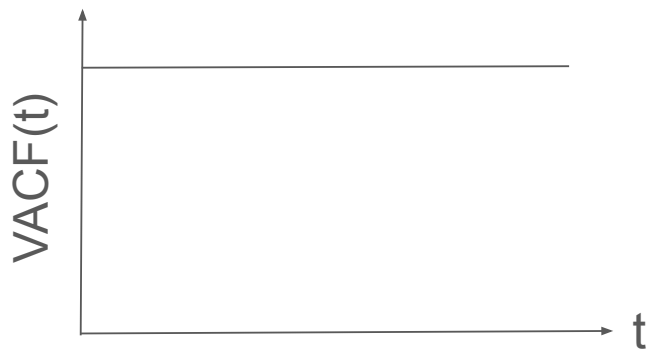
compute_vacf(vels):

choose τ

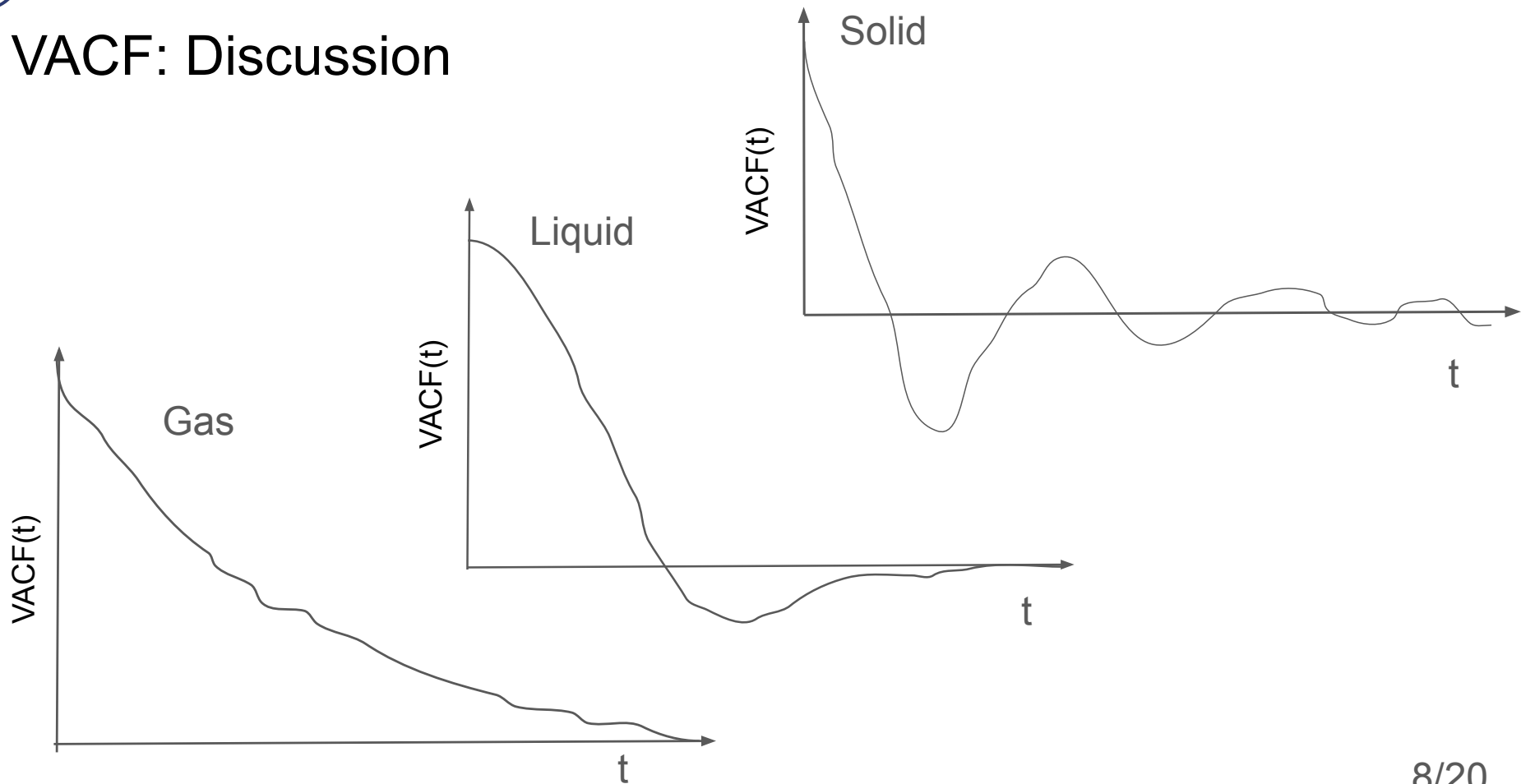
sum all possible
 $\mathbf{v}(t) \cdot \mathbf{v}(t+\tau)$

divide the sum by
the number of dot
products

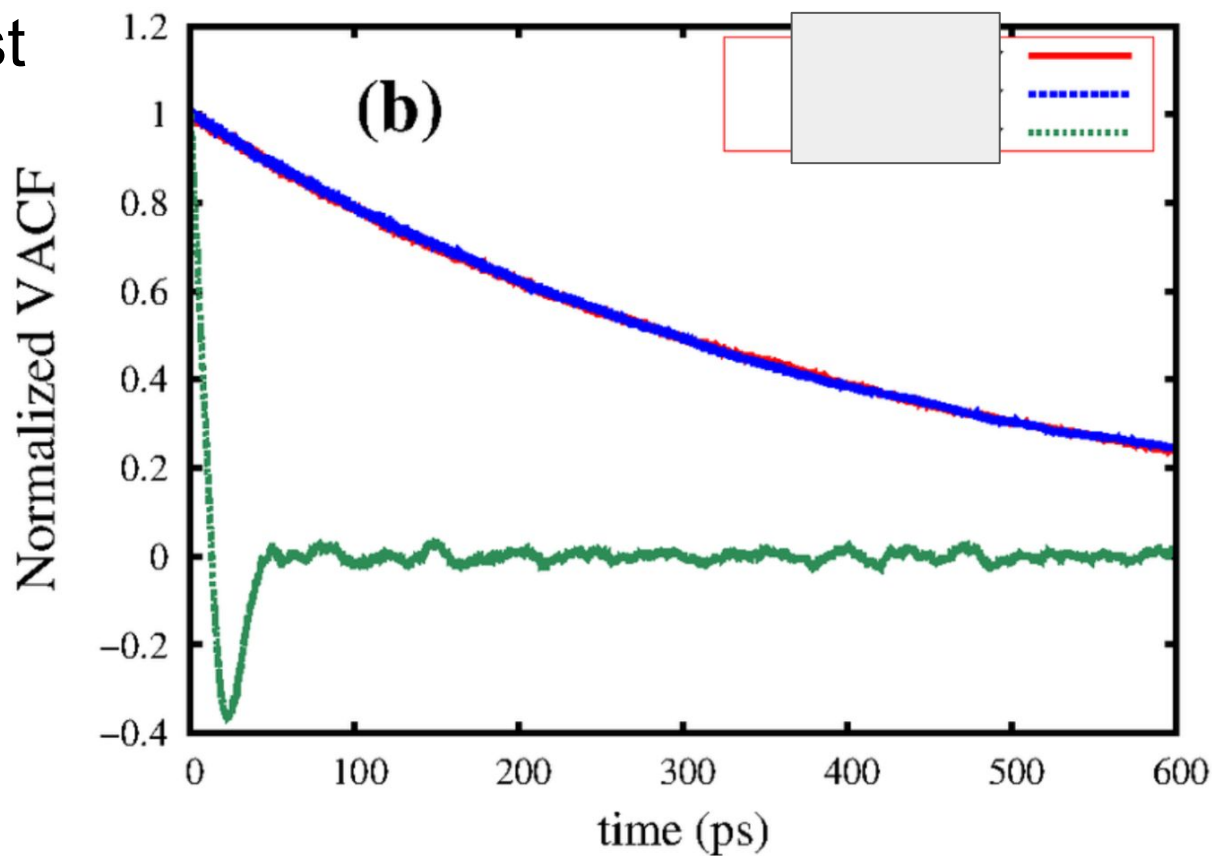
VACF: Discussion



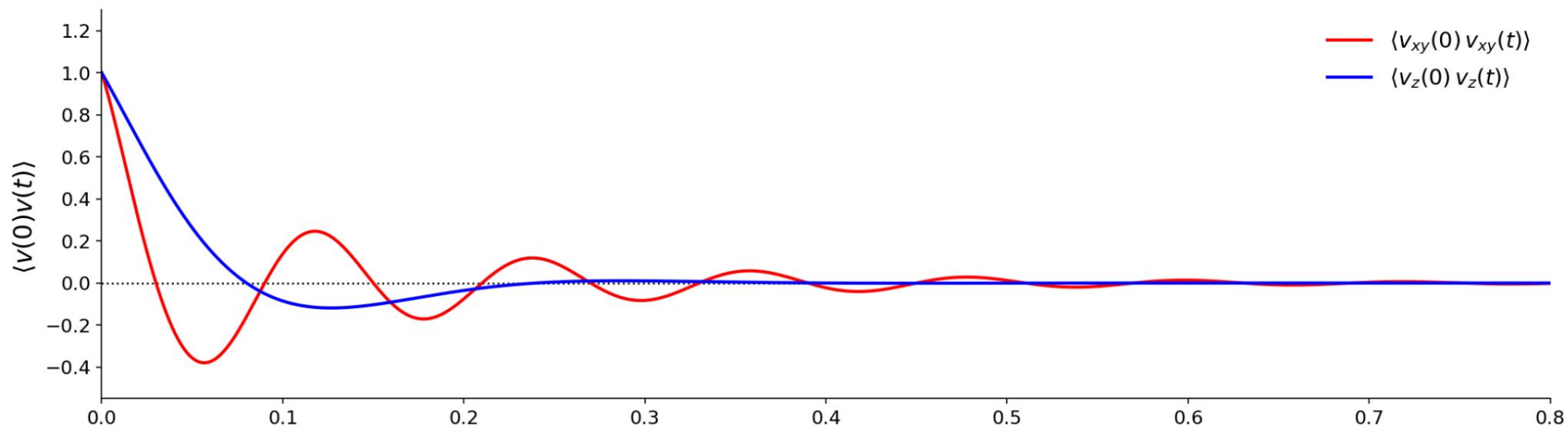
VACF: Discussion



VACF: Test

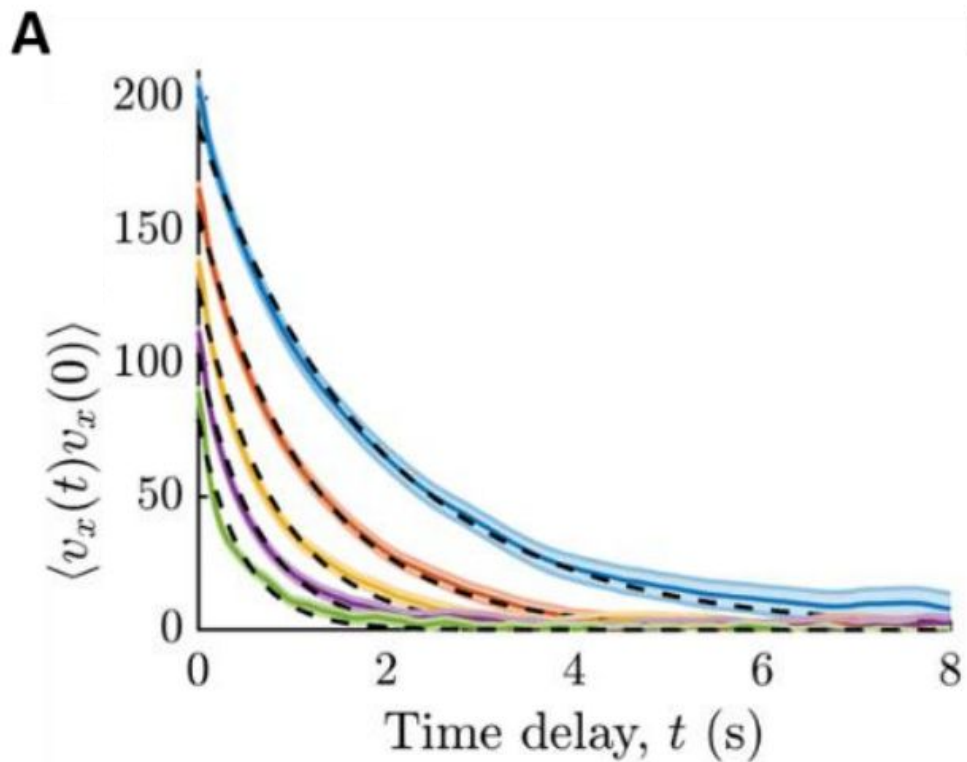


VACF: Test



Adapted from: Coretti, A., Baird, T., Vuilleumier, R., & Bonella, S. (2022). *The Journal of Chemical Physics*, 157(21).

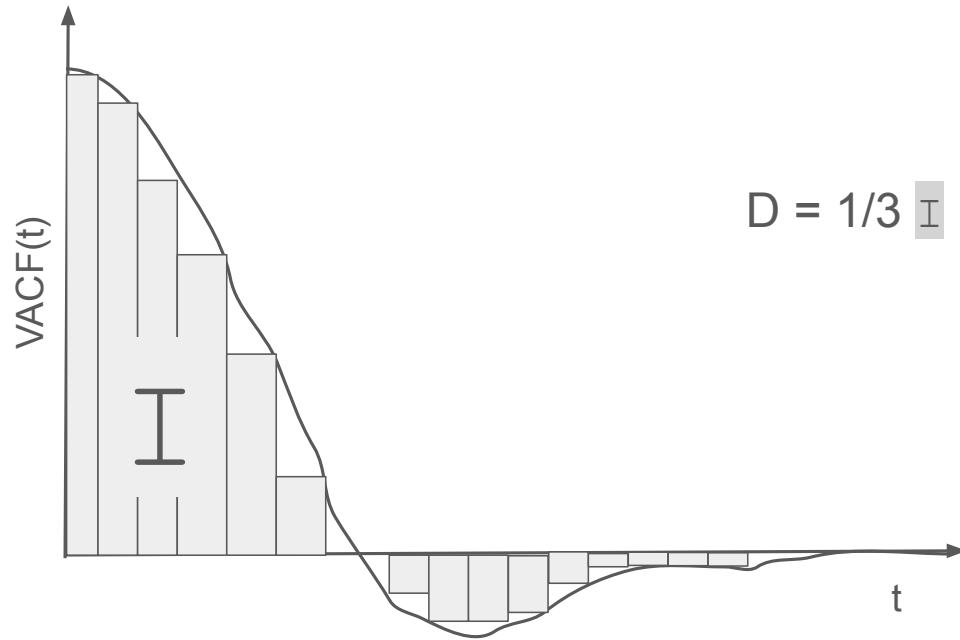
VACF: Test



Adapted from: Mattingly, H.H., Kamino, K., Machta, B.B. *et al. Nat. Phys.* 17, 1426–1431 (2021).

VACF: Discussion

- Diffusion



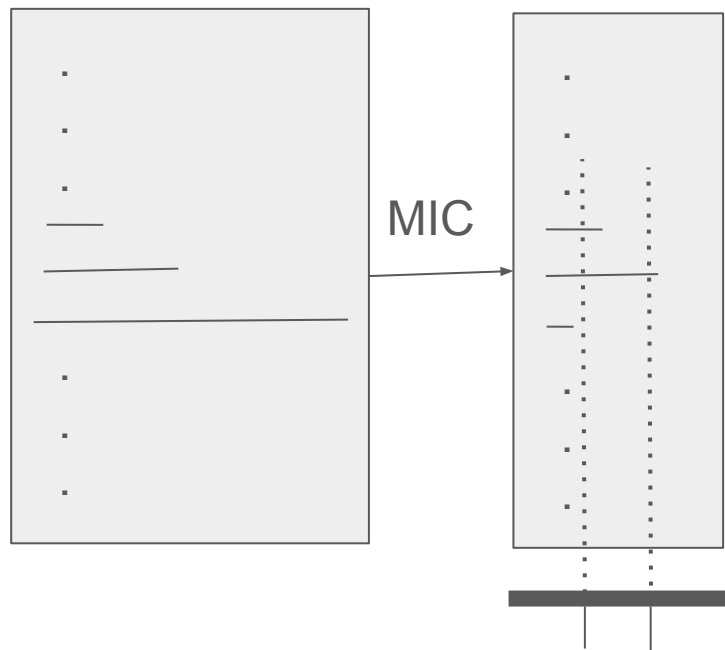
Diffusion

- Are there other ways to compute the diffusion coefficient?
- (a) time-integrating the velocity autocorrelation function $\langle \mathbf{v}_i(0) \cdot \mathbf{v}_i(t) \rangle$;
 - (b) cross-correlating the initial velocity with the displacement $\langle \mathbf{v}_i(0) \cdot \Delta \mathbf{r}_i(t) \rangle$;
 - (c) numerically time-differentiating the mean-squared displacement $\langle |\Delta \mathbf{r}_i(t)|^2 \rangle$;
 - (d) dividing the mean-squared displacement $\langle |\Delta \mathbf{r}_i(t)|^2 \rangle$ by the time.

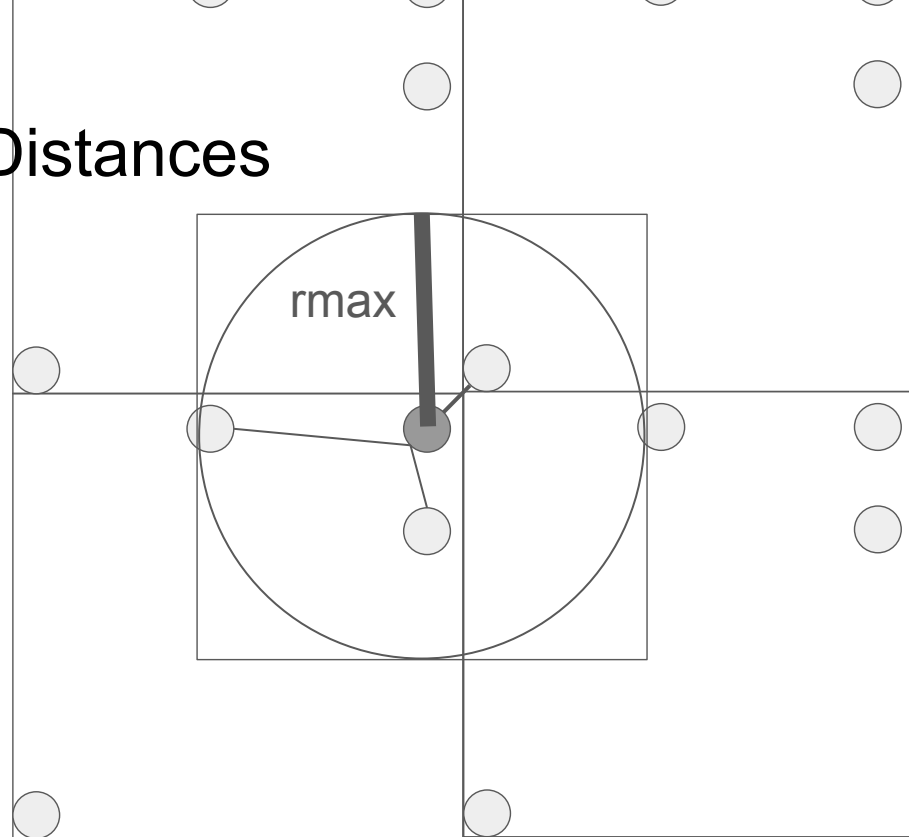
Radial Distribution Function ($g(r)$) : “Definition”

- Sitting on a typical atom, how does its local density at distance r compare to the bulk density $N_{\text{total}}/V_{\text{total}}$?
- Two Steps:
 1. Counting pair distances 
 2. Normalizing

$g(r)$: 1) Counting Pair Distances



pair_distance_counts = [# , # , #]



Nbins=3

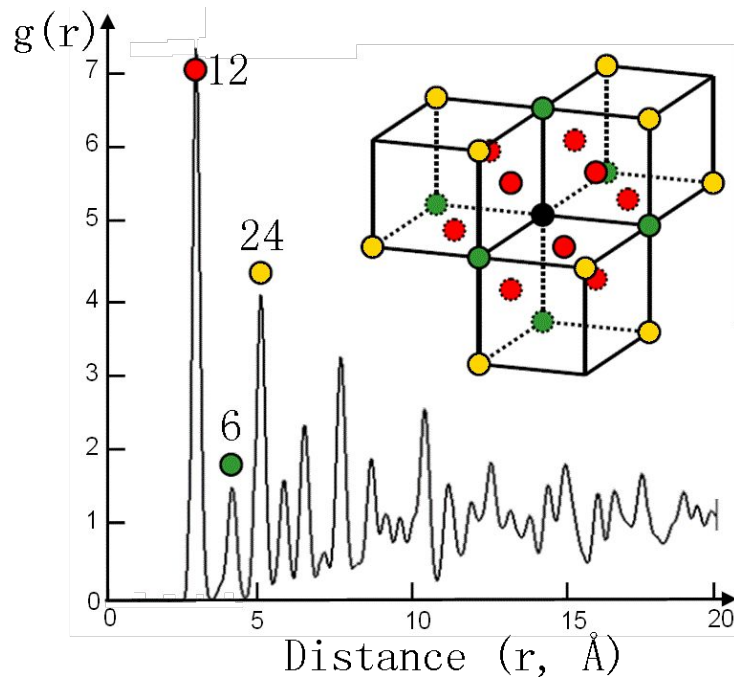
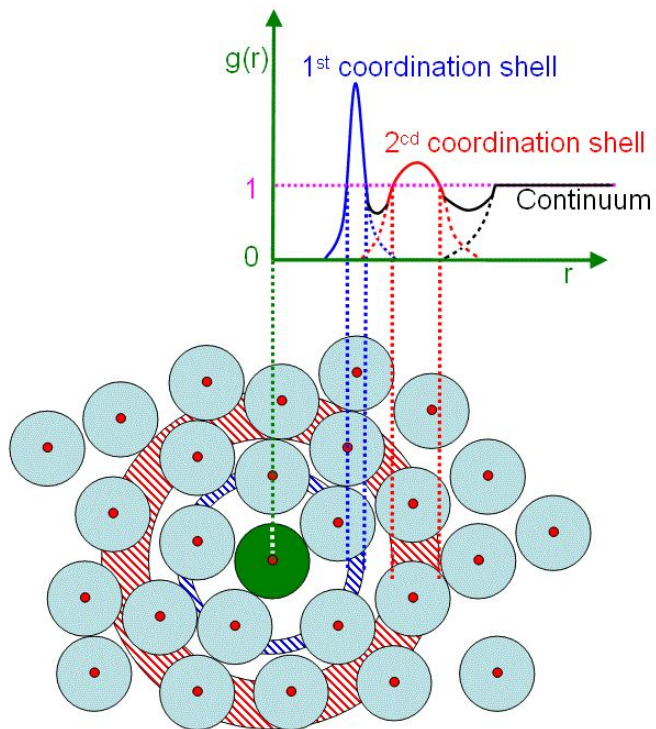
Lbin = $r_{max}/Nbins$

$g(r)$: 2) Normalization

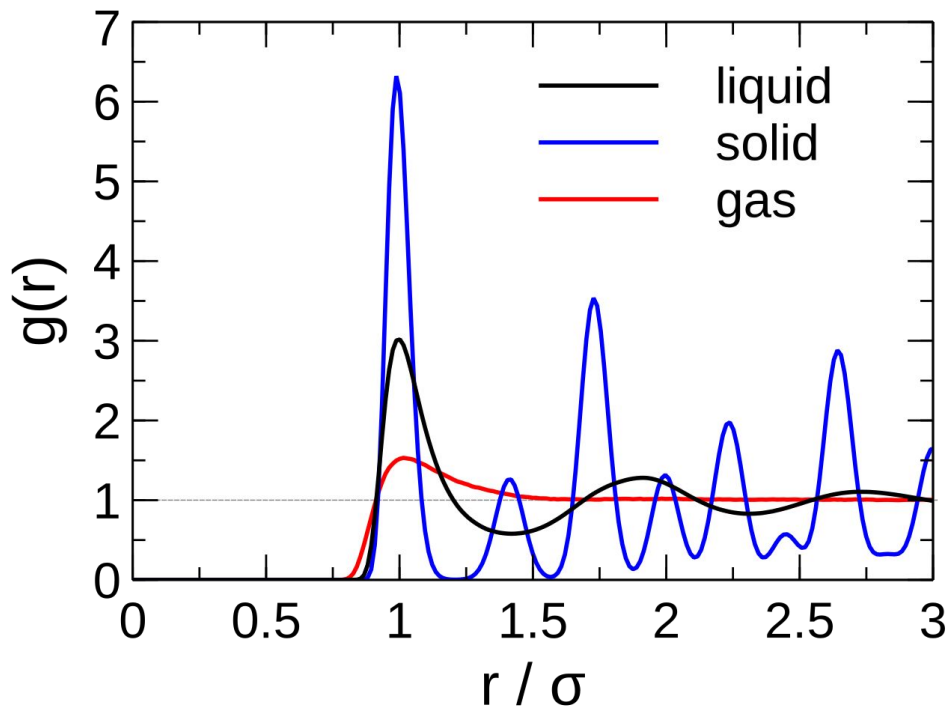
- Sitting on a typical atom, how does its local density at distance r compare to the bulk density $N_{\text{total}}/V_{\text{total}}$?

$$g(r) = \frac{\text{pair_distance_counts}}{\text{normalization ??}}$$

g(r): Discussion



g(r): Discussion



g(r): Discussion

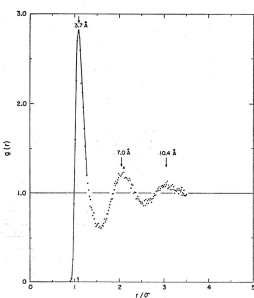
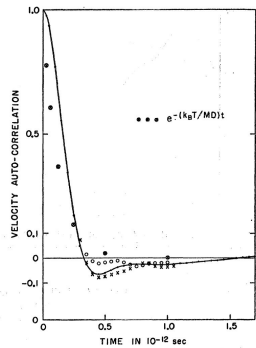
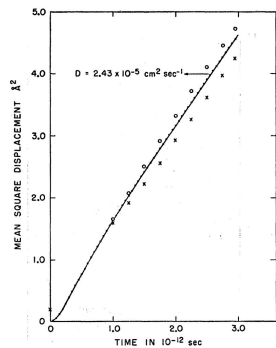
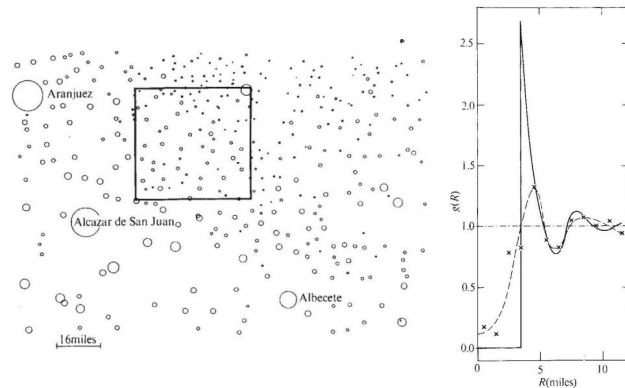
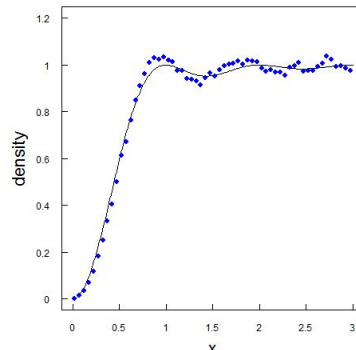


FIG. 2. Pair-correlation function obtained in this calculation at 34.4°K and 1.374 gm³. The Fourier transform of this function has peaks at $q = 6.8, 12.5, 18.5, 24.8$.



Glass, L., Tobler, W.. Nature 233, 67–68 (1971)

Rahman, A. (1964). *Physical review*, 136(2A), A405.



Montgomery, H. L. (1973). In *Proc. Symp. Pure Math* (Vol. 24, No. 1).

adapted form: https://en.wikipedia.org/wiki/Montgomery%27s_pair_correlation_conjecture#

g(r): Discussion

